



The Geometry of Generalization

Synthesizing Instance-Based and Model-Based Machine Learning

Two Extremes of Supervised Learning



Instance-Based Methods

- Represents learned knowledge as a collection of raw prototypes.
- Couched in the exact same language used to describe the training data.
- Derives predictions implicitly through similarity measures.

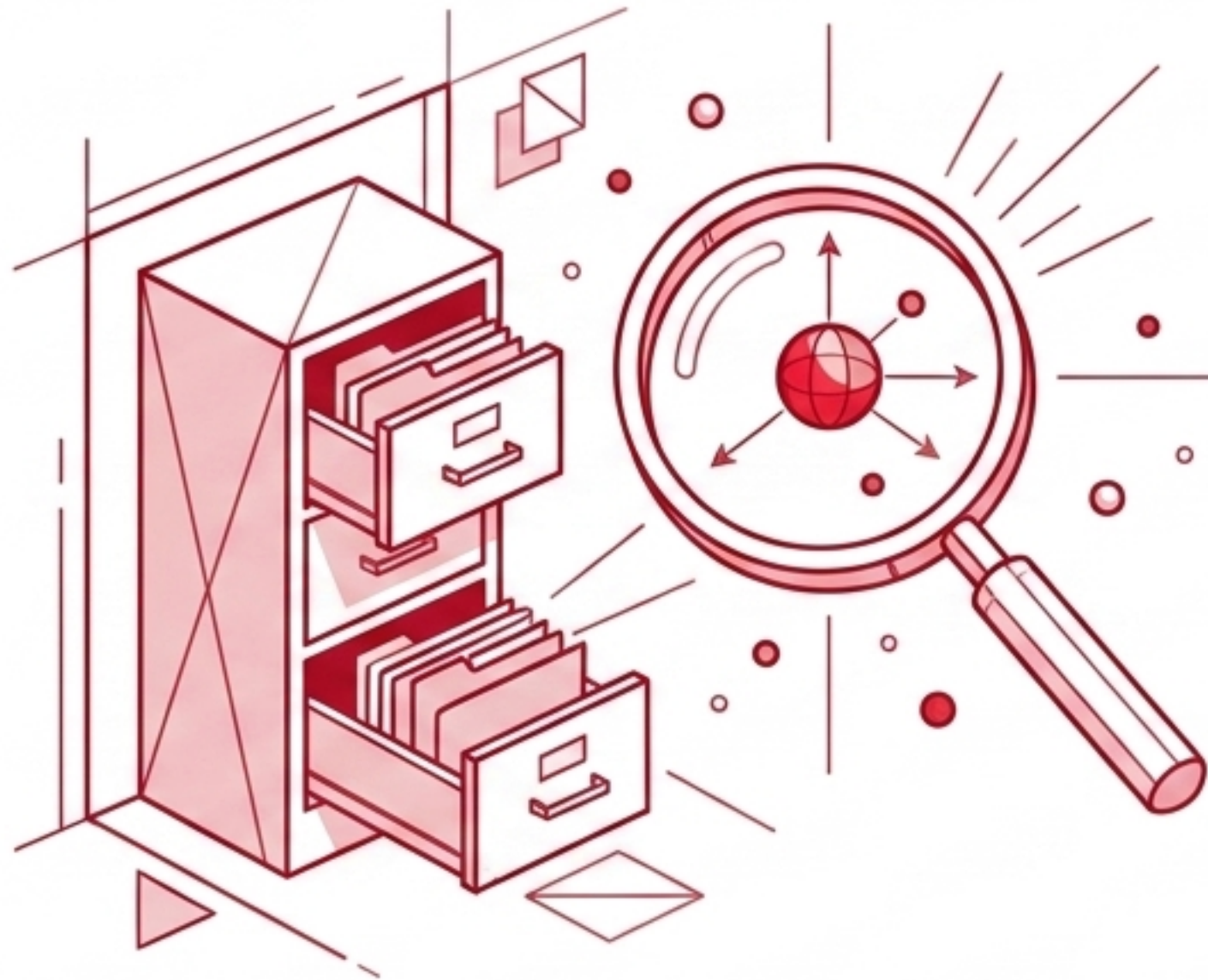


Model-Based Methods

- Represents learned knowledge in a richer, explicit theory language.
- Constructs overarching mathematical generalizations of the training cases.
- Derives predictions explicitly through an abstract model.

The Lazy Paradigm: Instance-Based Learning

Instead of estimating a target function once for the entire instance space, instance-based methods simply store the training examples.



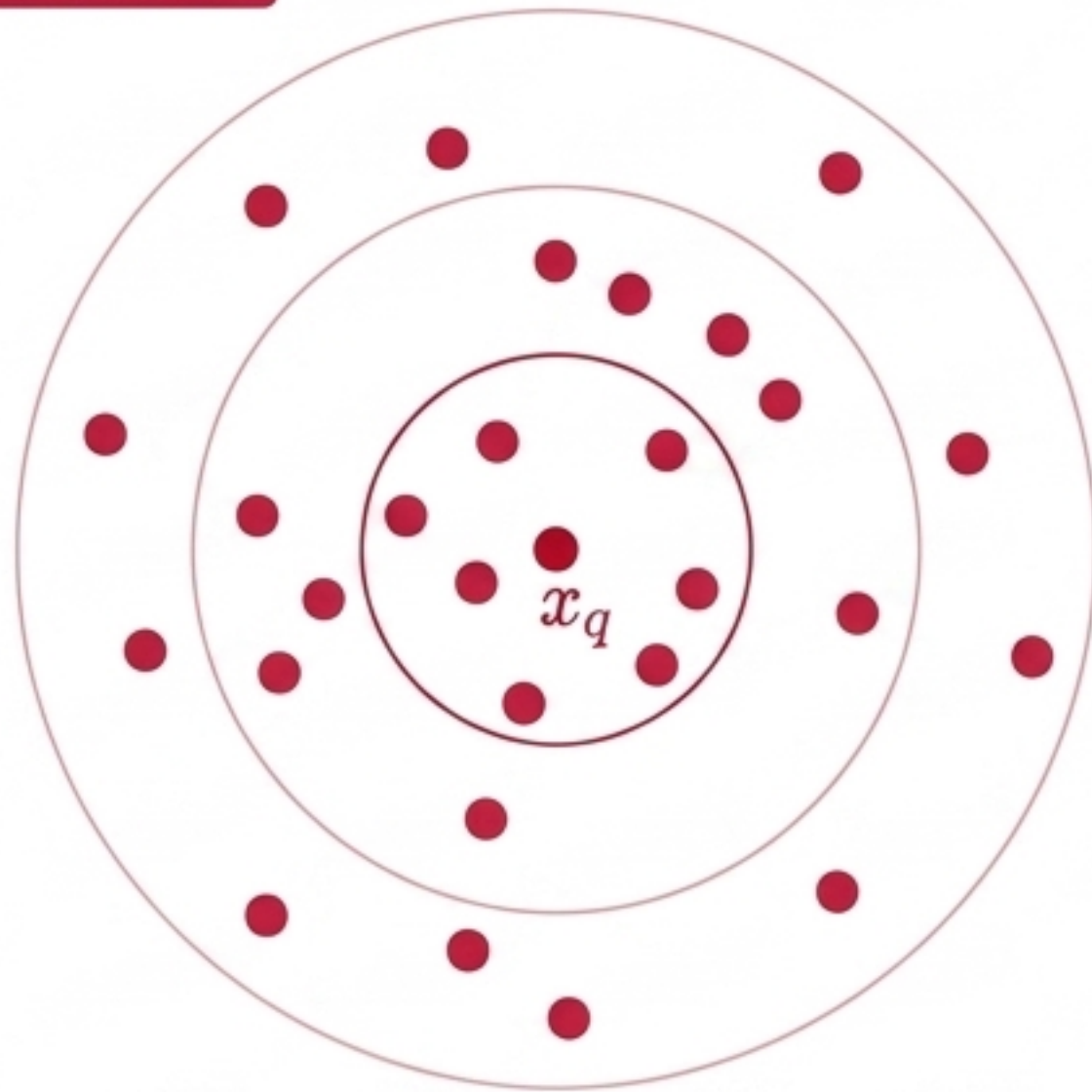
Core Characteristics:

- **Postponed Generalization:** Processing is delayed until a new, unseen instance must be classified.
- **Local Estimation:** The target function is estimated locally and differently for every single new query.
- **Information Retention:** Never loses nuance; every single data point is retained in memory.

Mechanics of Local Approximation

The k -Nearest Neighbor algorithm assumes all instances correspond to points in an n -dimensional space.

Crimson Red



Distance Metric:

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^n (x_{ir} - x_{jr})^2}$$

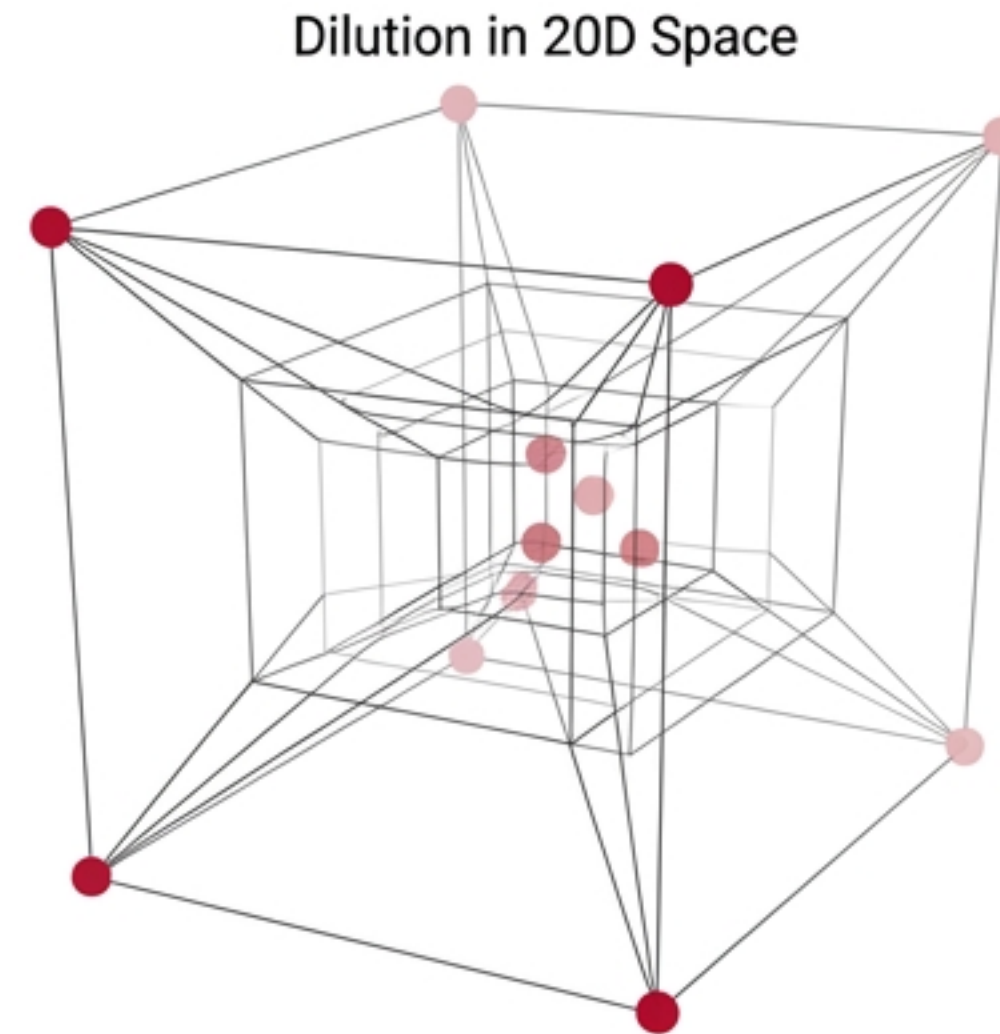
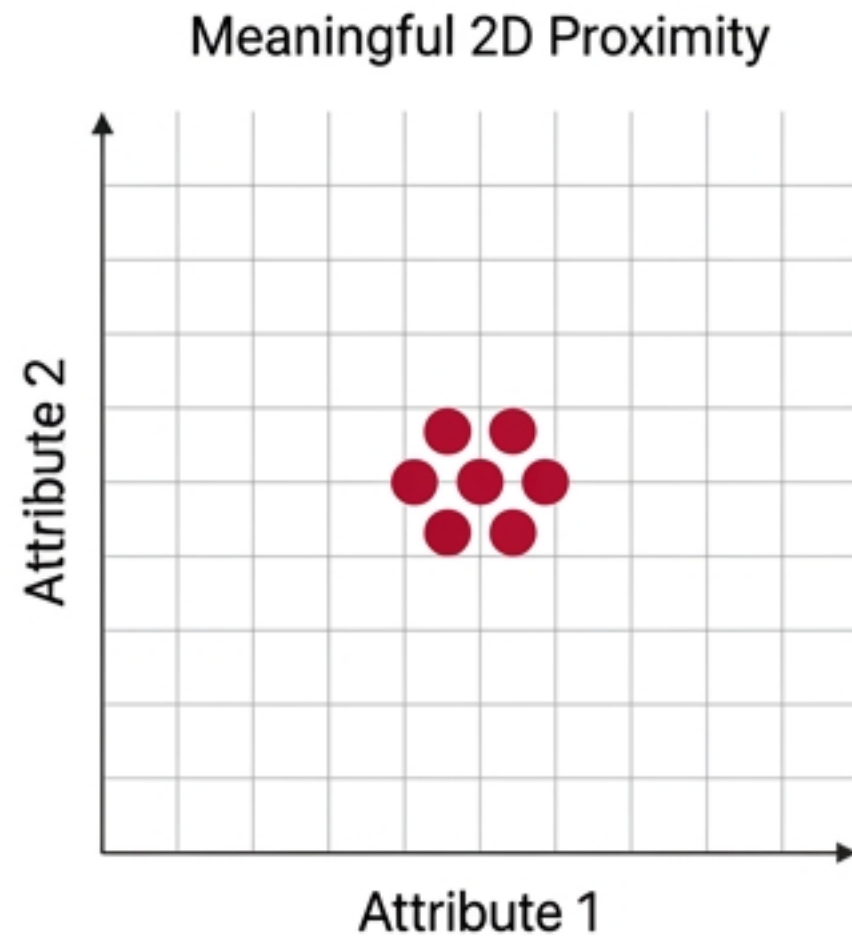
The Prediction Mechanism:

For a real-valued target, the algorithm calculates the mean of the f -values of the k nearest neighbors.

$$\hat{f}(x_q) \leftarrow \frac{\sum_{i=1}^k w_i f(x_i)}{\sum_{i=1}^k w_i}, \text{ where } w_i = \frac{1}{d(x_q, x_i)^2}$$

The Breaking Point: The Curse of Dimensionality

Instance-based methods falter when the task domain contains many irrelevant attributes.



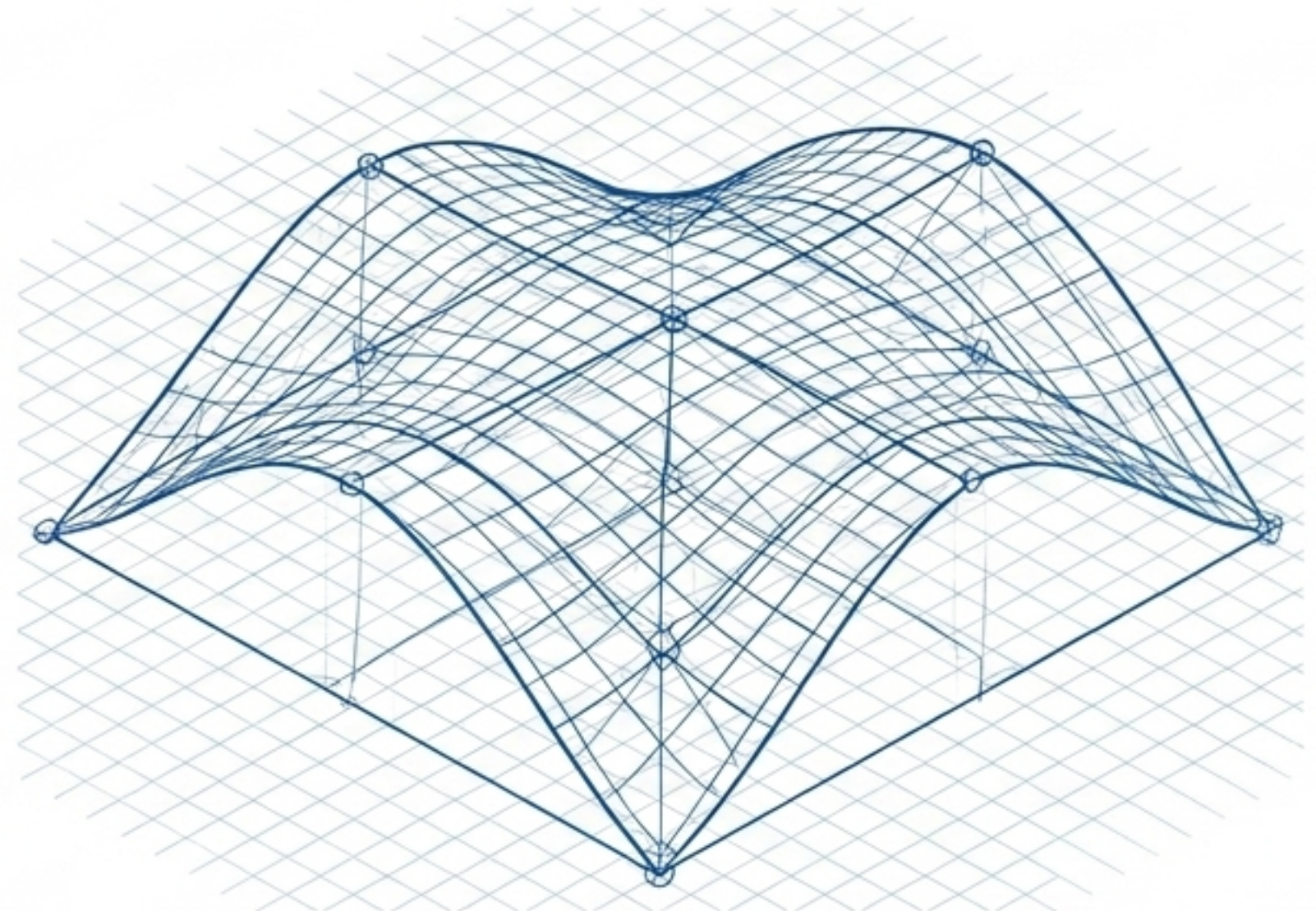
1. Imagine instances described by 20 attributes, where only 2 are actually relevant to the target function.
2. In this high-dimensional space, the nearest neighbor is easily misled.
3. Because the algorithm treats all dimensions equally by default, the true local neighborhood is drowned out by the noise of irrelevant axes.

The Eager Paradigm: Model-Based Learning

Eager learning methods construct an explicit, **global** approximation of the target function before seeing a single query.

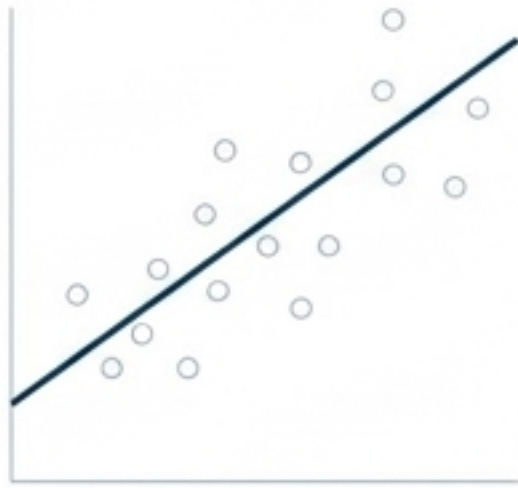
Core Characteristics:

- **Upfront Generalization:** Heavy processing occurs during the training phase to build the model.
- **Abstract Representation:** The model relies on mathematical parameters (weights, coefficients) rather than raw data instances.
- **Robustness to Noise:** By seeking a global minimum error, these models can effectively ignore irrelevant attributes and smooth over noisy data points.



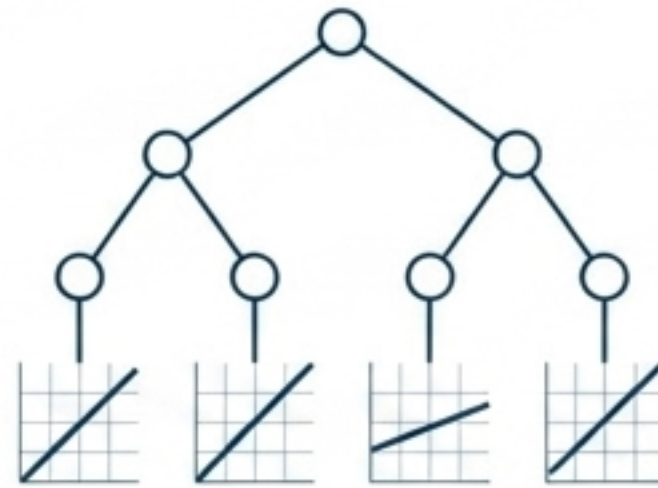
Constructing the Global Surface

Three standard approaches to building predictive, continuous-value models.



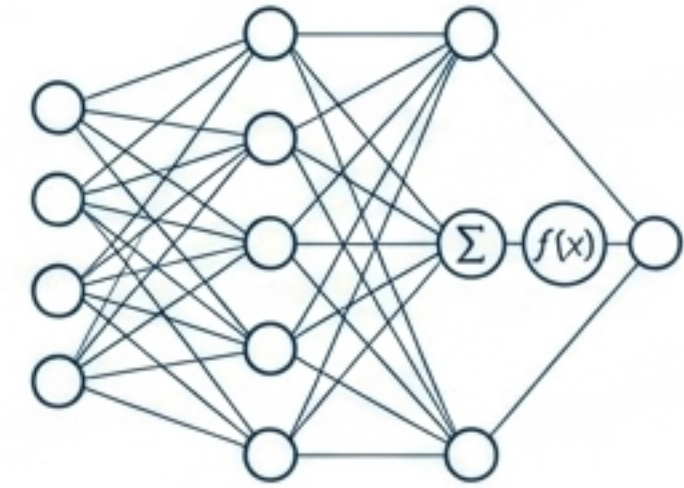
Multivariate Linear Regression

Finds coefficients to minimize the sum of squared differences. Intelligible, but assumes a strict linear relationship across the entire space.



Model Trees (m5)

Uses recursive partitioning to build a piecewise linear model. Splits the training cases to minimize intra-subset variation, placing linear models at the leaves.

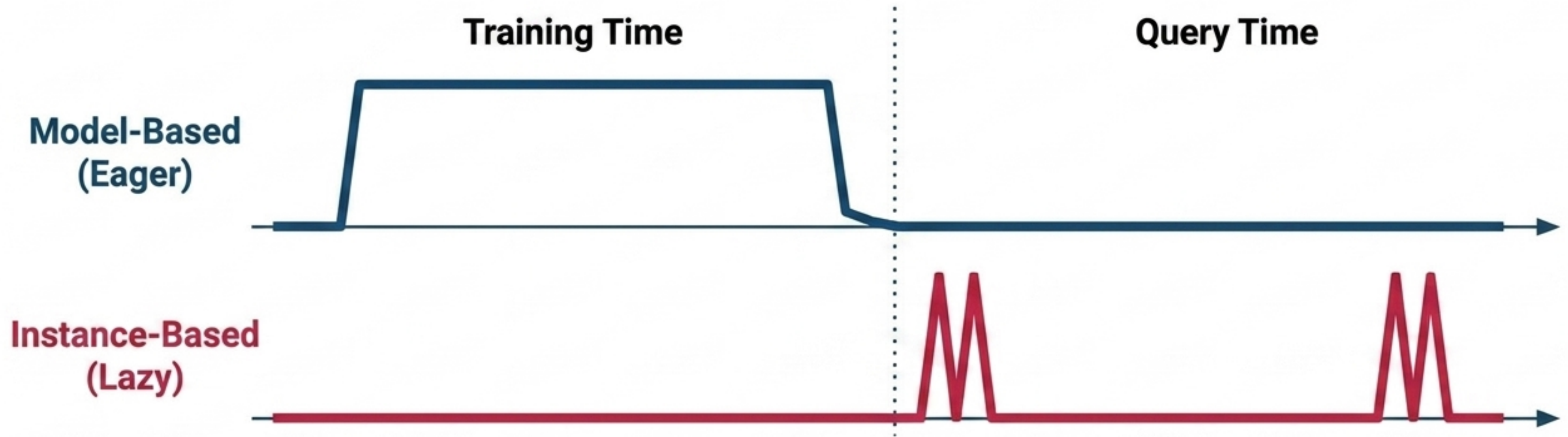


Neural Networks

Utilizes hidden layers and logistic functions to convert combined inputs into outputs, minimizing error across the global dataset via conjugate gradient methods.

The Generalization Timeline

The defining distinction between the paradigms is exactly when the computational work of generalization occurs.



- **Model-Based (Eager):** Slow to train, lightning-fast to query. The mathematical surface is pre-calculated.

- **Instance-Based (Lazy):** Zero training time, slow to query. The local geography must be mapped anew for every single prediction.

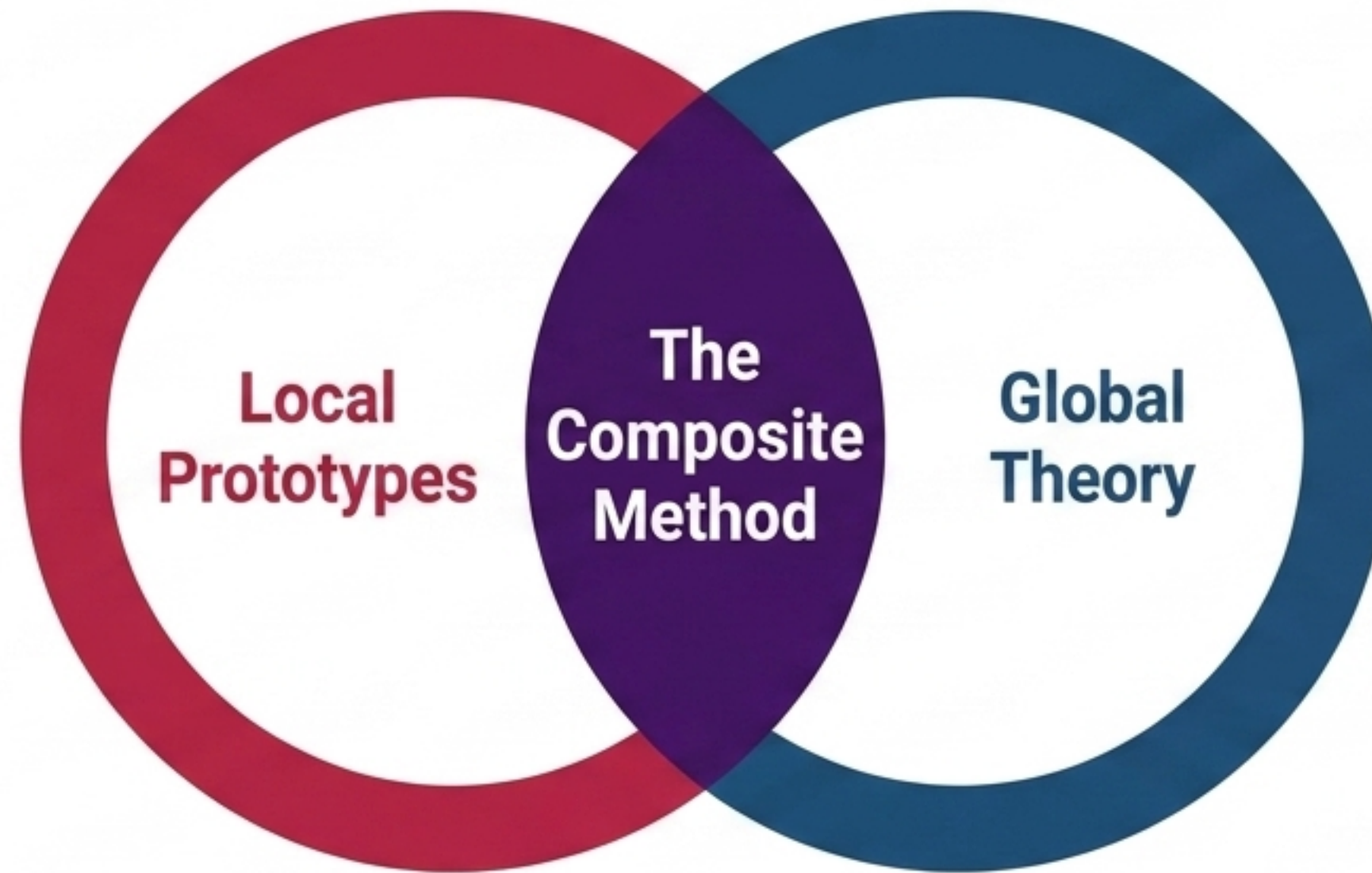
The Master Diagnostic Matrix

Dimension	Instance-Based (Lazy)	Model-Based (Eager)
Approximation Scope	Localized to query	Global across dataset
Training Speed	Instant (✓)	Computationally heavy (✗)
Query Speed	Computationally heavy (✗)	Instant (✓)
Irrelevant Attributes	Easily misled (✗)	Highly resilient (✓)
Data Retention	Lossless	Lossy (Abstracted)
Representative Methods	kNN, Case-Based Reasoning	Linear Regression, Neural Nets

The Synthesis: A Composite Methodology

Pure instance-based methods ignore structural trends.

Pure model-based methods ignore local anomalies.



The theoretical ideal is a composite method: utilizing the exact same training data to simultaneously provide a global structural model and localized prototype information. Both forms of knowledge must be brought to bear on the unseen case.

The Mechanism of Correction

How do we adjust a local prototype using a global model?

Let $\mathbf{U}$ be the unseen query. Let $\mathbf{P}_i$ be its nearest neighbor prototype. Let $\mathbf{M}$ be the global model.

Step 1: The Local Baseline

The nearest neighbor's actual value is $\mathbf{V}(\mathbf{P}_i)$.

Because $\mathbf{P}_i$ is similar to $\mathbf{U}$, $\mathbf{V}(\mathbf{P}_i)$ is our starting guess.

Step 2: The Global Delta

The model predicts the structural difference between the two points to be:

$$\mathbf{M}(\mathbf{P}_i) - \mathbf{M}(\mathbf{U}).$$

Step 3: The Adjusted Prediction

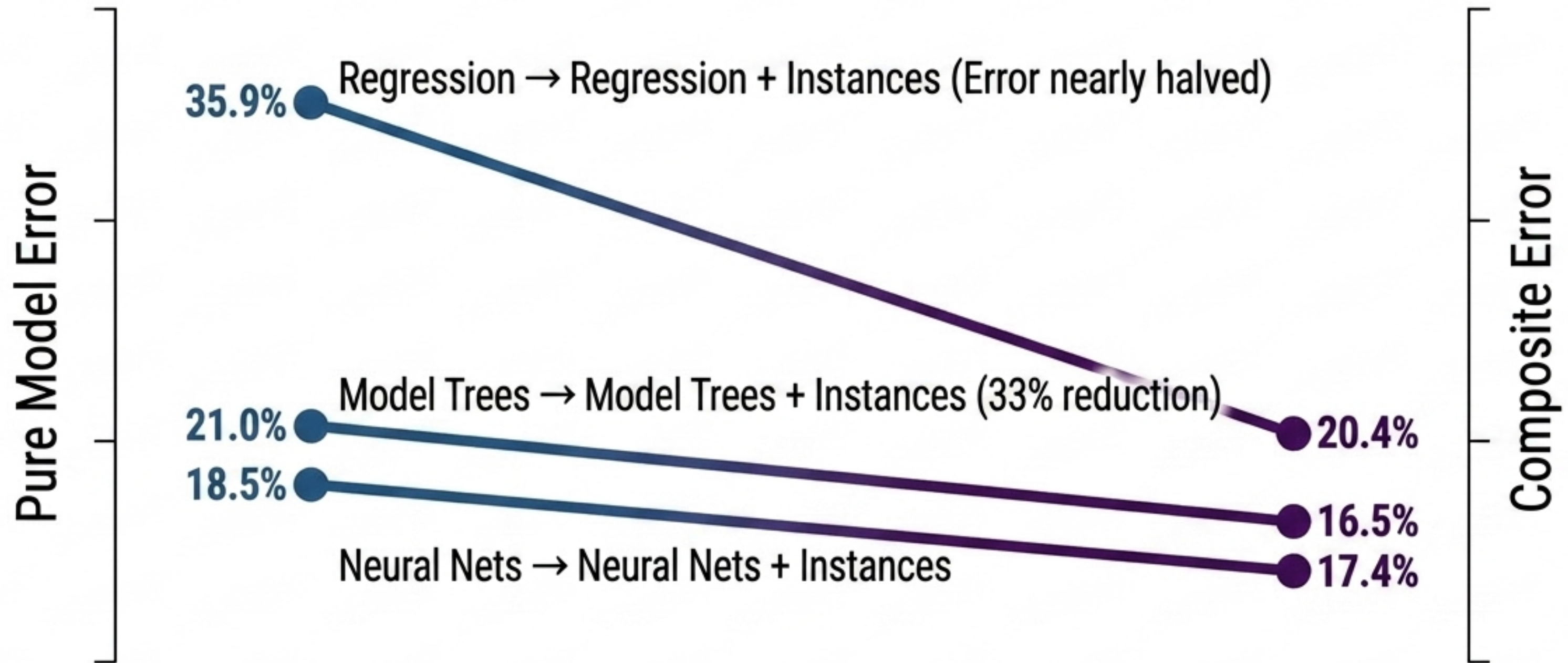
We subtract the expected global difference from the local actual value.

$$\mathbf{V}(\mathbf{P}_i) - (\mathbf{M}(\mathbf{P}_i) - \mathbf{M}(\mathbf{U}))$$

If the model is correct, this adjustment yields a far more precise predictor than the raw prototype alone.

Empirical Validation

Over a representative collection of datasets, composite methods routinely produce better predictions than either pure approach in isolation.

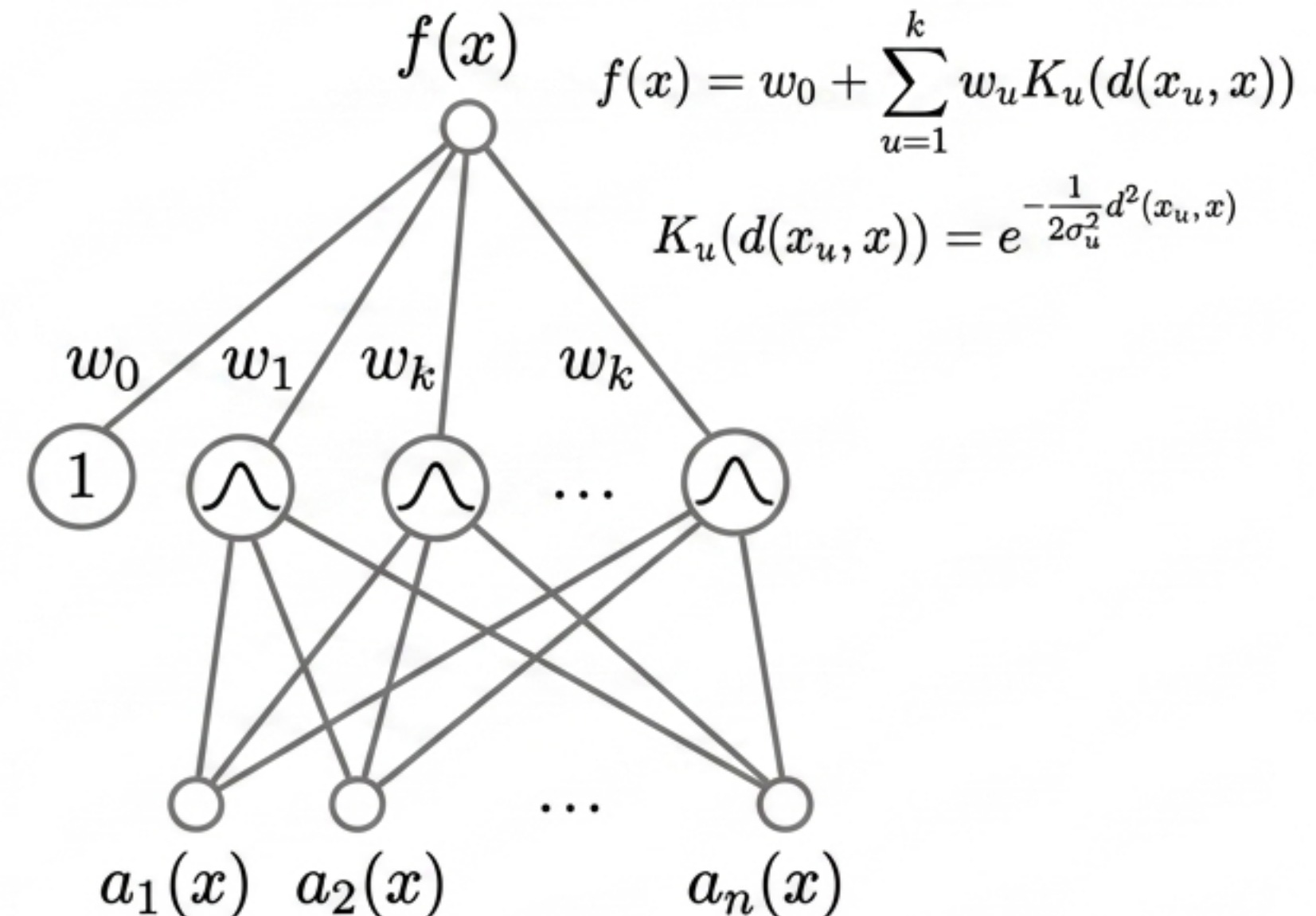


(Data sourced from 10-way cross-validation on datasets including housing, cpu, and servomechanisms).

An Alternative Synthesis: Radial Basis Functions

Function approximation bridges the gap by building a global hypothesis exclusively out of localized components.

- **The Architecture:** An artificial neural network where each hidden unit produces an activation determined by a Gaussian kernel function.
- **Local Anchors:** The Gaussian function is centered at a specific training instance (x_u). Its activation drops to near zero unless the input query (x) is near x_u .
- **Global Output:** The final output unit produces a linear combination of these purely local hidden unit activations, forming a complete global surface.



The Unified Blueprint

The architecture of optimal predictive learning requires both localized memory and generalized theory.

- 1. Acknowledge the Trade-offs:** Lazy learning retains perfect nuance but breaks under high-dimensionality. Eager learning resists noise but smooths over localized reality.
- 2. Combine the Toolsets:** Use global models to explicitly calculate the structural differences between points, and use local instances to anchor those predictions in empirical reality.
- 3. Synthesize for Precision:** Whether through composite formulas or localized kernels (RBFs), the future of predictive accuracy lies in the synthesis of the local and the global.